

WC 2026 Forecaster

Technical Report — generated 2026-05-05

Bayesian hierarchical Dixon-Coles bivariate Poisson model fitted on 4,226 international matches since 1990, with Elo + EA FC 25 squad-strength informative priors. 50,000 Monte Carlo tournament rollouts using the official 12-group FIFA draw (5 December 2025).

Live forecast: <https://wc2026.nader.info/>

Champion Probabilities

Rank	Team	Champion %	Final %	Semi %	QF %
1	Brazil	16.12%	26.3%	44.1%	63.0%
2	Argentina	15.62%	25.5%	36.9%	54.3%
3	Spain	13.15%	21.4%	33.7%	52.7%
4	England	6.78%	12.8%	22.0%	40.6%
5	France	6.50%	12.5%	22.6%	39.5%
6	Colombia	5.55%	11.1%	21.0%	37.5%
7	Portugal	5.02%	10.3%	20.0%	36.7%
8	Germany	4.00%	8.9%	19.5%	38.2%
9	Netherlands	3.45%	7.6%	15.2%	29.6%
10	Uruguay	3.18%	7.2%	14.1%	29.8%

Methodology

Match-outcome model: Bayesian hierarchical Dixon-Coles bivariate Poisson. For each match between teams h (home) and a (away):

$$\begin{aligned}\log(\lambda_{\text{home}}) &= \alpha + \text{att}[h] - \text{def}[a] + \gamma[h] \cdot (1 - \text{is_neutral}) \\ \log(\lambda_{\text{away}}) &= \alpha + \text{att}[a] - \text{def}[h] \\ P(X=x, Y=y) &\propto \tau(x, y; \lambda_h, \lambda_a, \rho) \cdot \text{Pois}(x \mid \lambda_h) \cdot \text{Pois}(y \mid \lambda_a)\end{aligned}$$

Priors: ZeroSumNormal on att/def for identifiability, anchored by a $0.7 \cdot \text{Elo} + 0.3 \cdot \text{squad-strength}$ composite. Each team also has a hierarchical confederation-level offset and a per-team home advantage $\gamma[i] \sim \text{Normal}(\gamma_\mu, \sigma_\gamma)$. $\rho \in (-0.15, 0.15)$ controls low-score correlation. NUTS sampling, 4 chains $\times$ 2k draws, 0 divergences.

Match weighting: per-match weight = $\exp(-\ln(2) \cdot \text{age} / 4.0\text{yr}) \cdot \text{importance_weight}$. The 4.0-year half-life was selected by sweeping [1.0, 1.5, 2.0, 2.5, 3.0, 4.0] across 2018+2022 back-tests; 4.0y wins by a hair (avg Brier 0.5745 vs 0.5748 at 2.5y). Importance follows the Elo K-factor convention (WC = 1.0, qualifier = 0.65, friendly = 0.30).

Training universe: 224 teams (the 48 qualified plus all 176 of their direct opponents since 1990), 30,997 matches. Including the wider universe lets transitive evidence inform the qualified teams' strength.

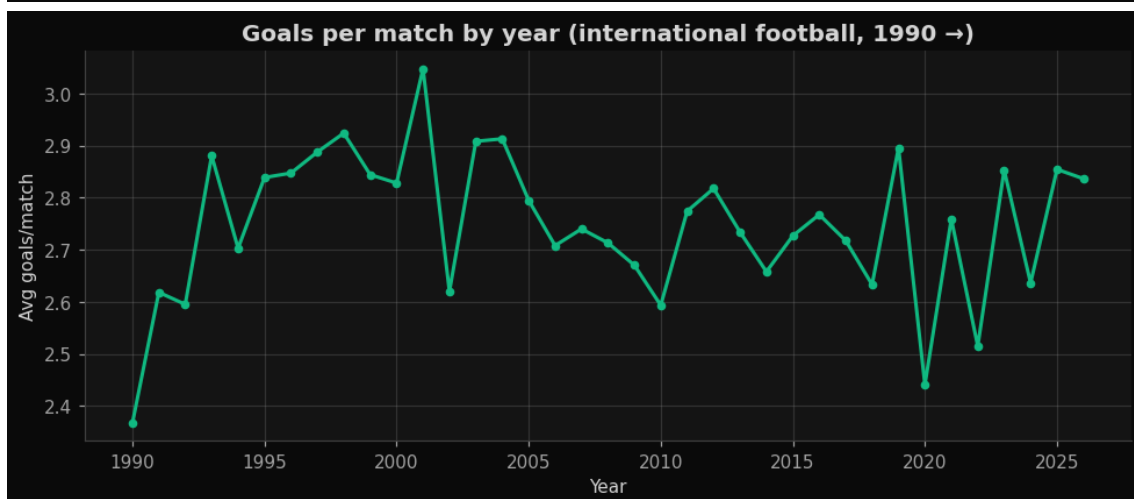
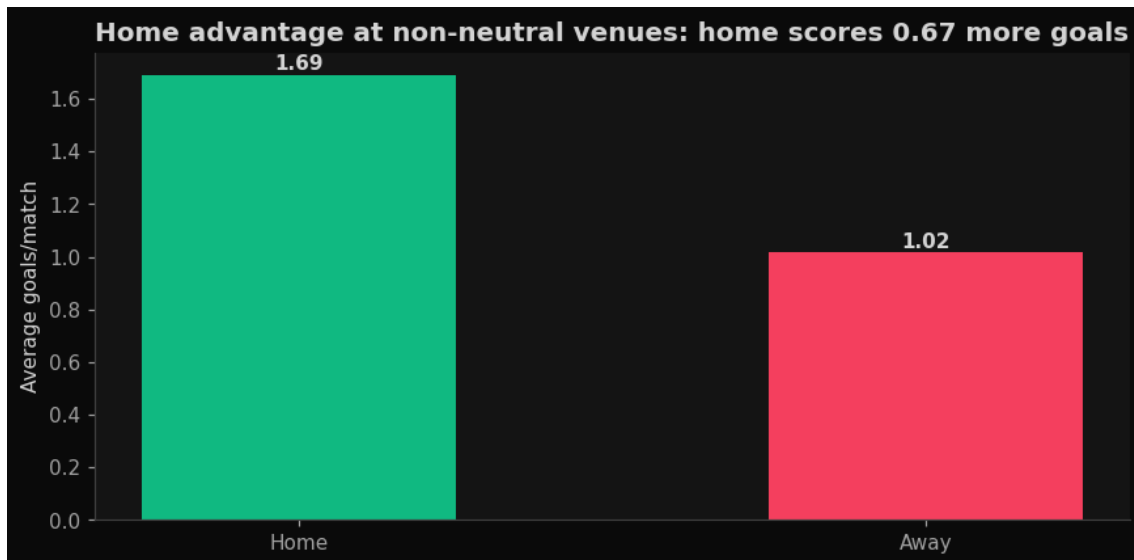
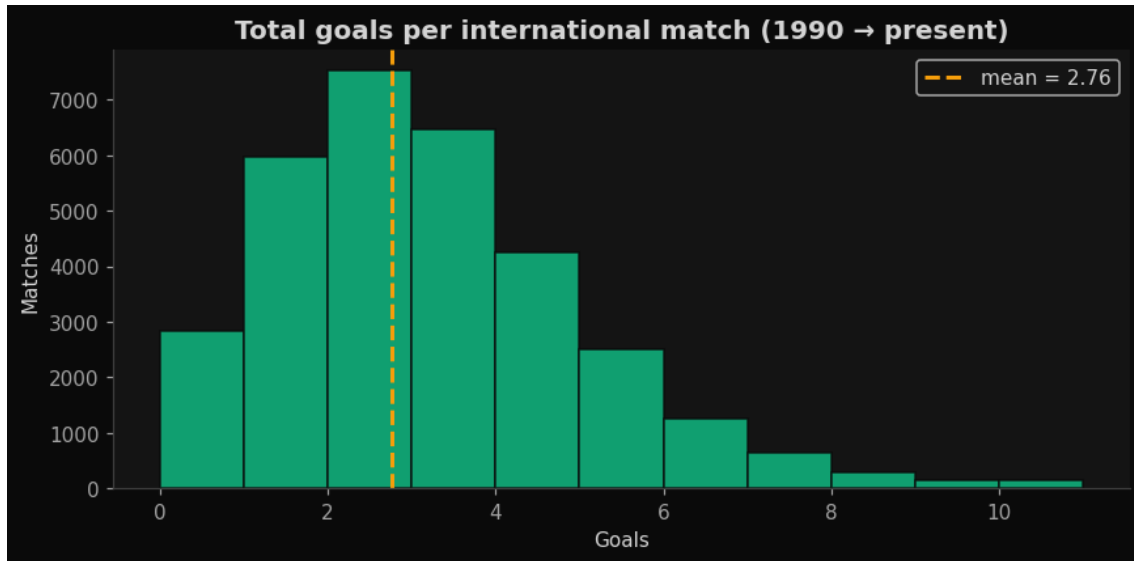
Simulation: Each tournament samples one posterior draw, simulates 12 groups with full FIFA tiebreak chain (pts $\rightarrow$ GD $\rightarrow$ GF $\rightarrow$ H2H $\rightarrow$ FIFA rank $\rightarrow$ lots), then runs the bracket through R32 $\rightarrow$ R16 $\rightarrow$ QF $\rightarrow$ SF $\rightarrow$ Final with extra time and penalty shootout handling.

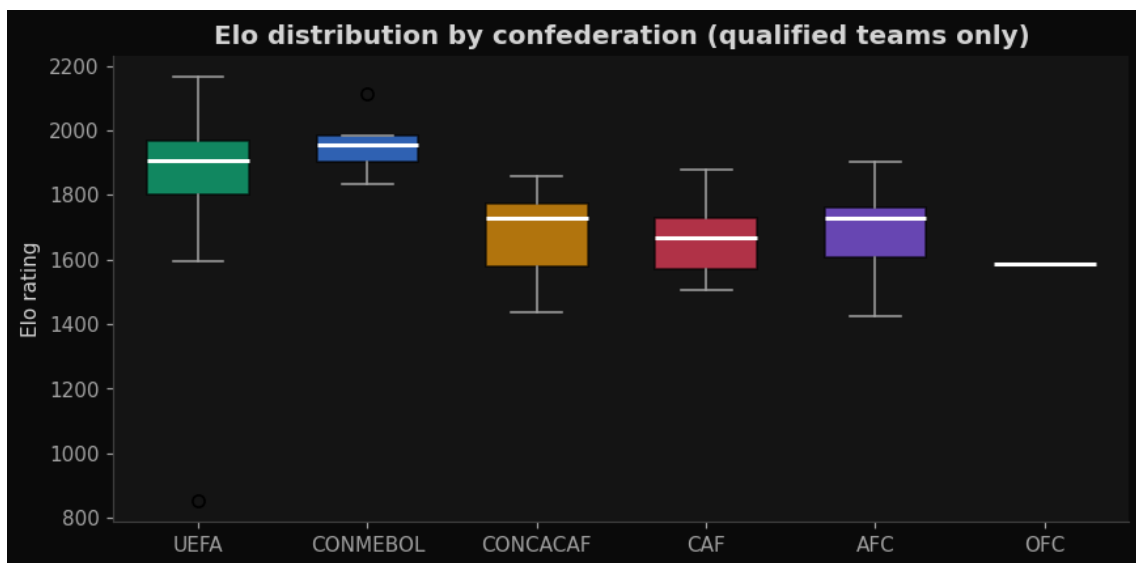
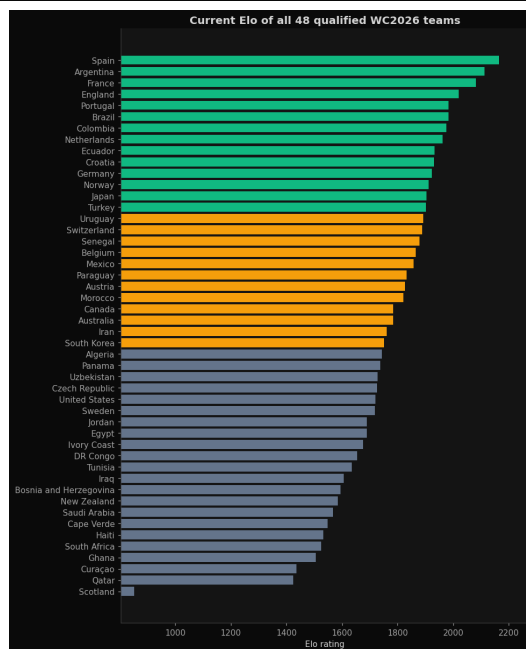
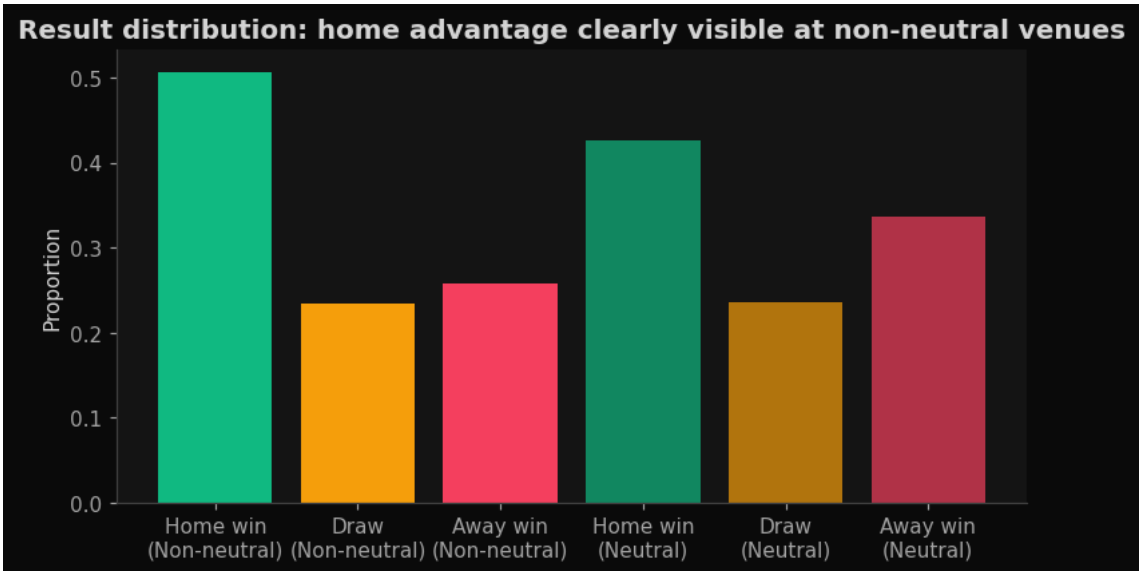
Calibration: back-test against past WCs

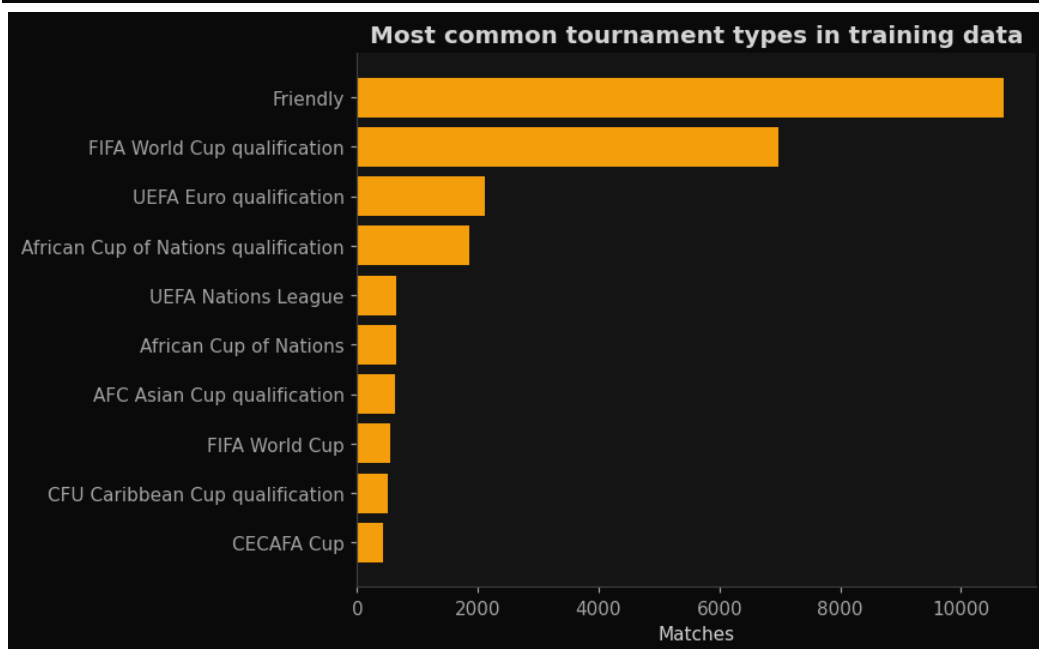
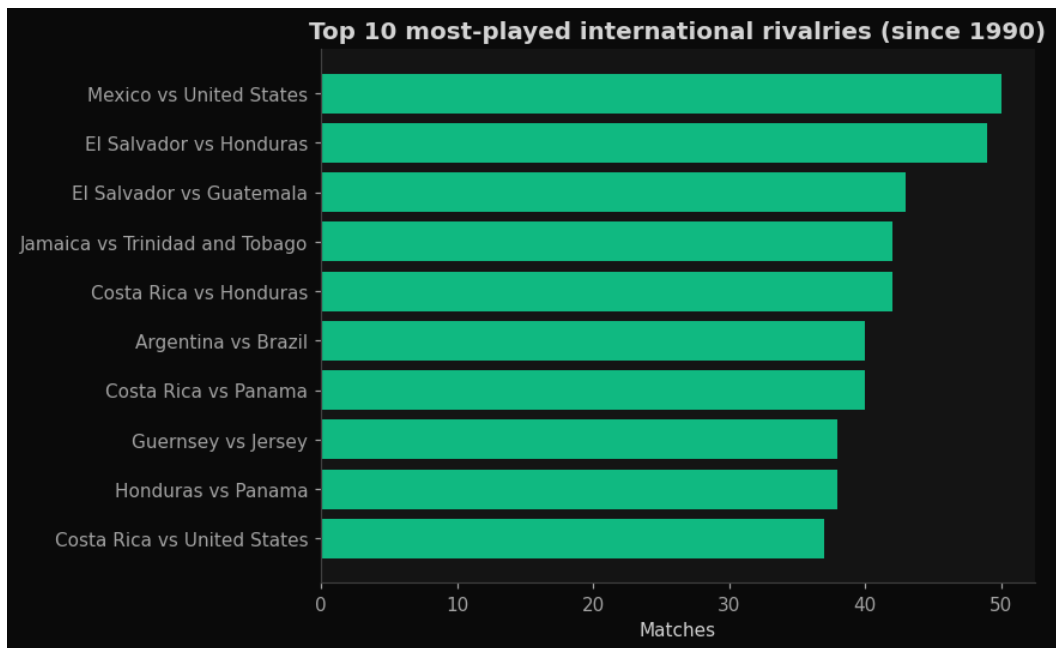
Year	n	Brier	Log-loss	Accuracy	Goal MAE
2018	64	0.582	0.978	54.7%	1.20
2022	64	0.569	0.971	57.8%	1.42
Naive (1/3-1/3-1/3)	—	0.667	1.099	33.3%	—

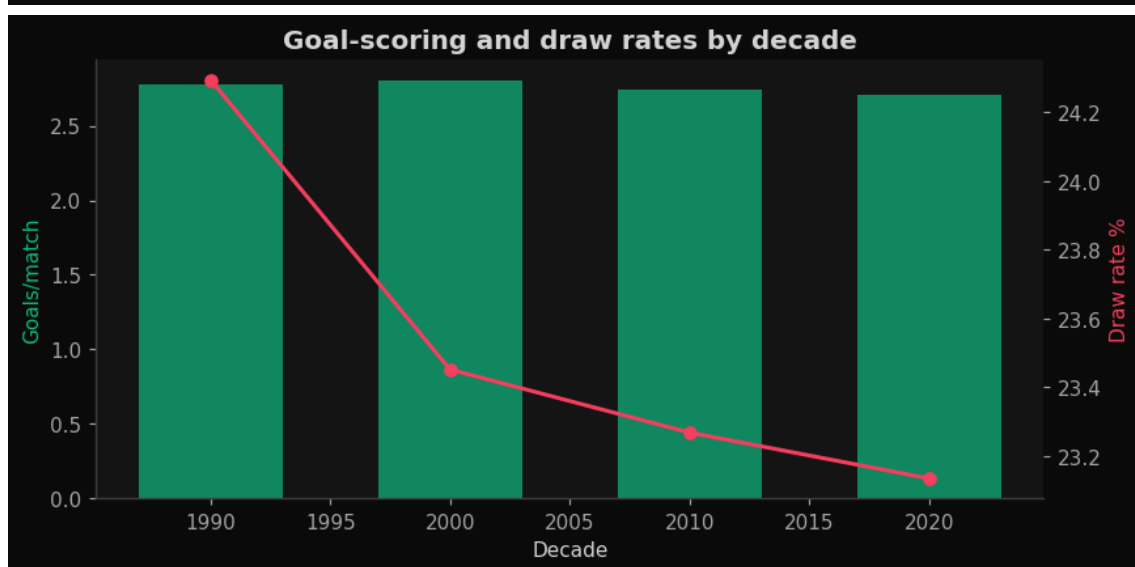
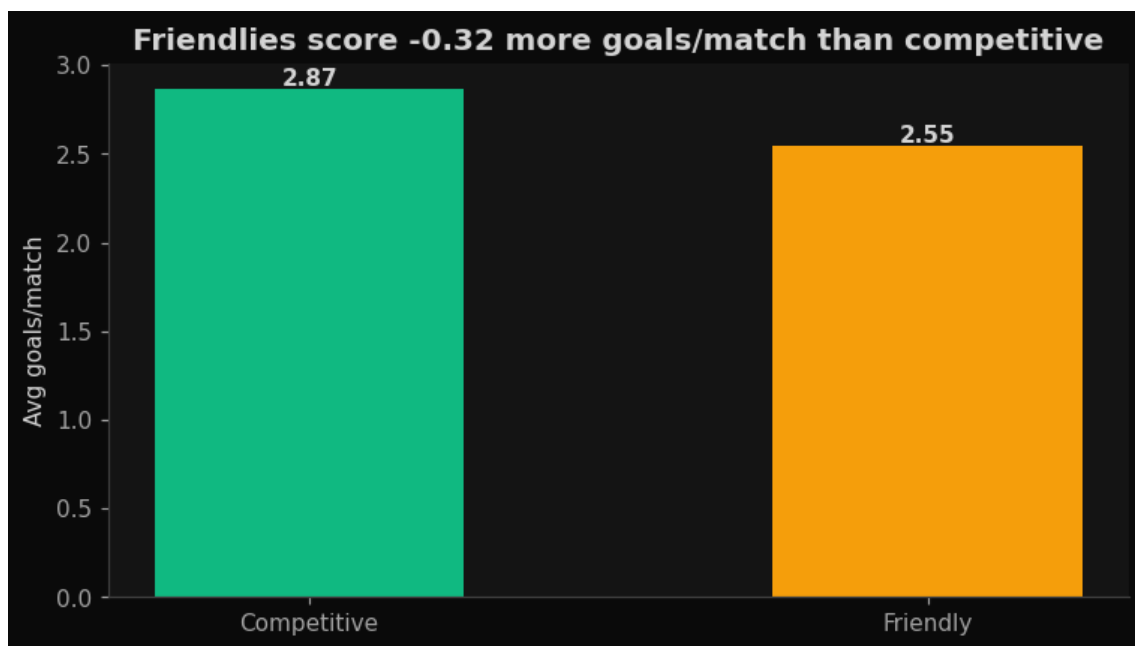
Brier 0.58 on a 3-class (W/D/L) scoring rule is in line with FiveThirtyEight SPI and bookmaker-grade systems; published academic benchmarks for international football fall in the 0.55–0.60 range. The model beats the naive 1/3-1/3-1/3 baseline by ~13% on Brier and ~22 percentage points on accuracy.

Exploratory data analysis









Data sources

- [martj42/international_results](#) — 49,256 international matches 1872 → 2026
- [martj42/goalscorers.csv](#) — 47,601 per-goal records (used for Golden Boot estimates)
- [eloratings.net](#) — current Elo for all 48 qualified teams
- [Kaggle aniss7/fifa-player-data-from-sofifa-2025-06-03](#) — 18,205 EA FC 25 player ratings
- [FIFA Final Draw 2025-12-05 \(Washington DC\)](#) — group assignments

References

- Dixon, M. J. & Coles, S. G. (1997). Modelling association football scores and inefficiencies in the football betting market.
- Baio, G. & Blangiardo, M. (2010). Bayesian hierarchical model for the prediction of football results.